

Blockchain-Based Supply Chain Financing Platform - Group 13

1. Environment:

- Solidity version: **0.8.20**
- Platform: **Remix VM (Cancun)**
- Accounts used:
 - Account 1 (Verifier/Admin)
 - Account 2 (Exporter)
 - Account 3 (Lender)

2. Test Case 1

Register Document

- Action: Account 2 calls registerDocument("INV001", 1 ether)
- Expected Result:
 - A new document is created
 - documentId = 1
 - status = Registered (0)
- Evidence: Figure 1 (DocumentRegistered event)

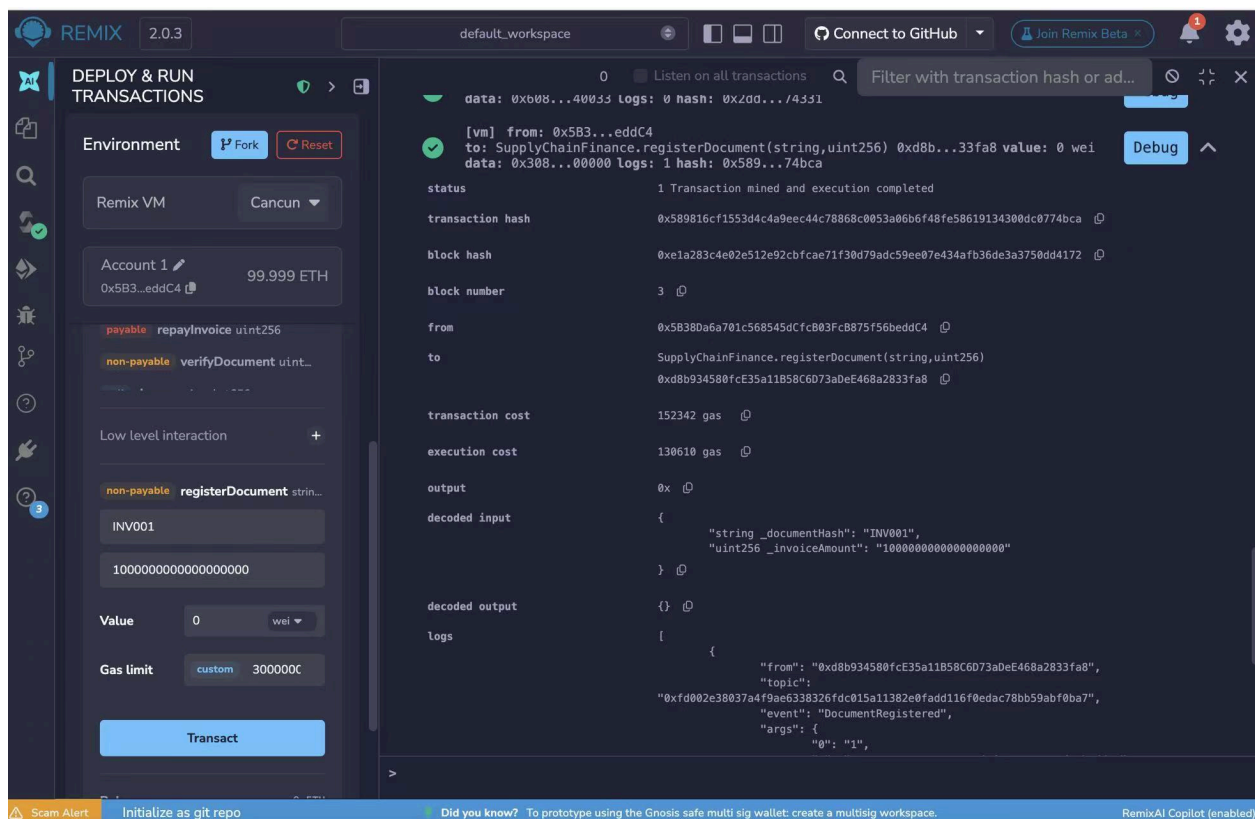


Figure 1: Register Document

3. Test Case 2

Verify Document

- Action: Account 1 calls verifyDocument(1)
- Expected Result:
 - Document status changes from Registered (0) to Verified (1)
- Evidence: Figure 2 (DocumentVerified event)

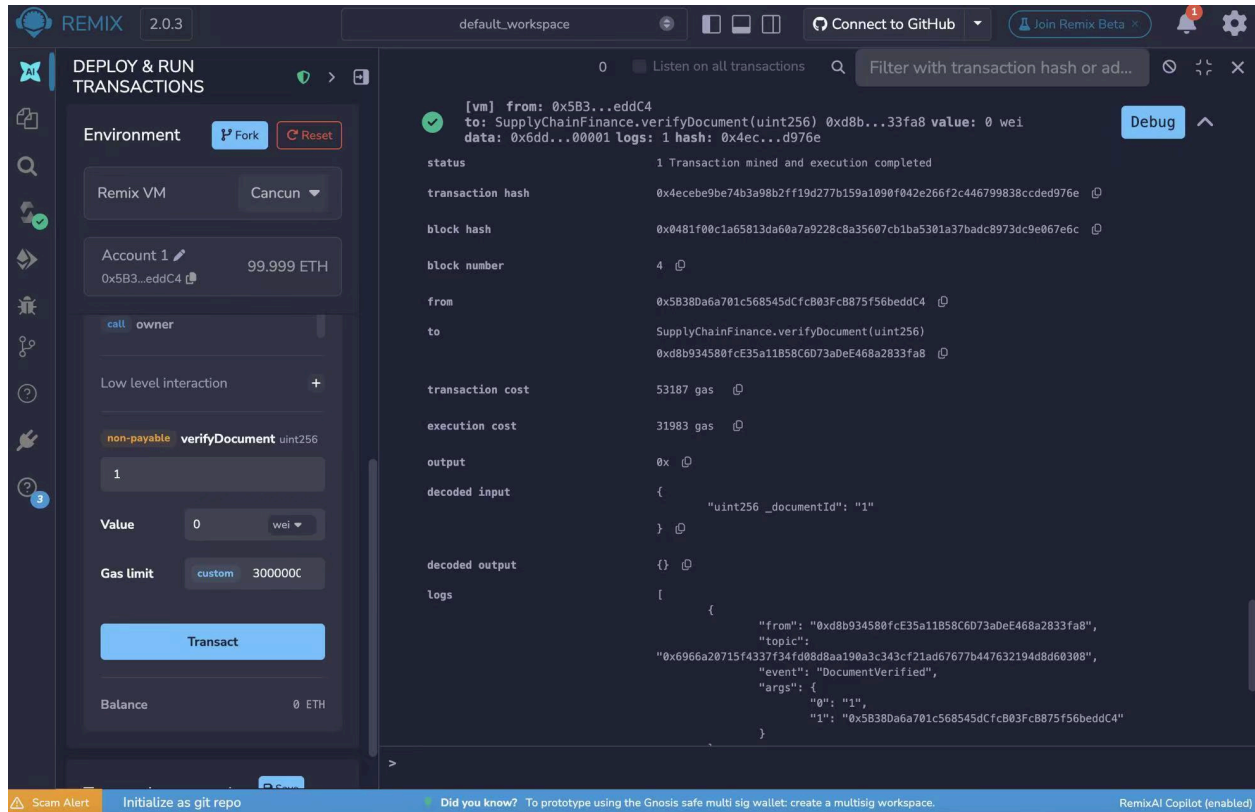


Figure 2: Verified Document

4. Test Case 3

Fund Invoice

- Action: Account 3 calls fundInvoice(1) with 1 ether
- Expected Result:
 - Document status changes to Funded (2)
 - Lender address is recorded
 - Funds are transferred to exporter
- Evidence: Figure 3 (InvoiceFunded event)

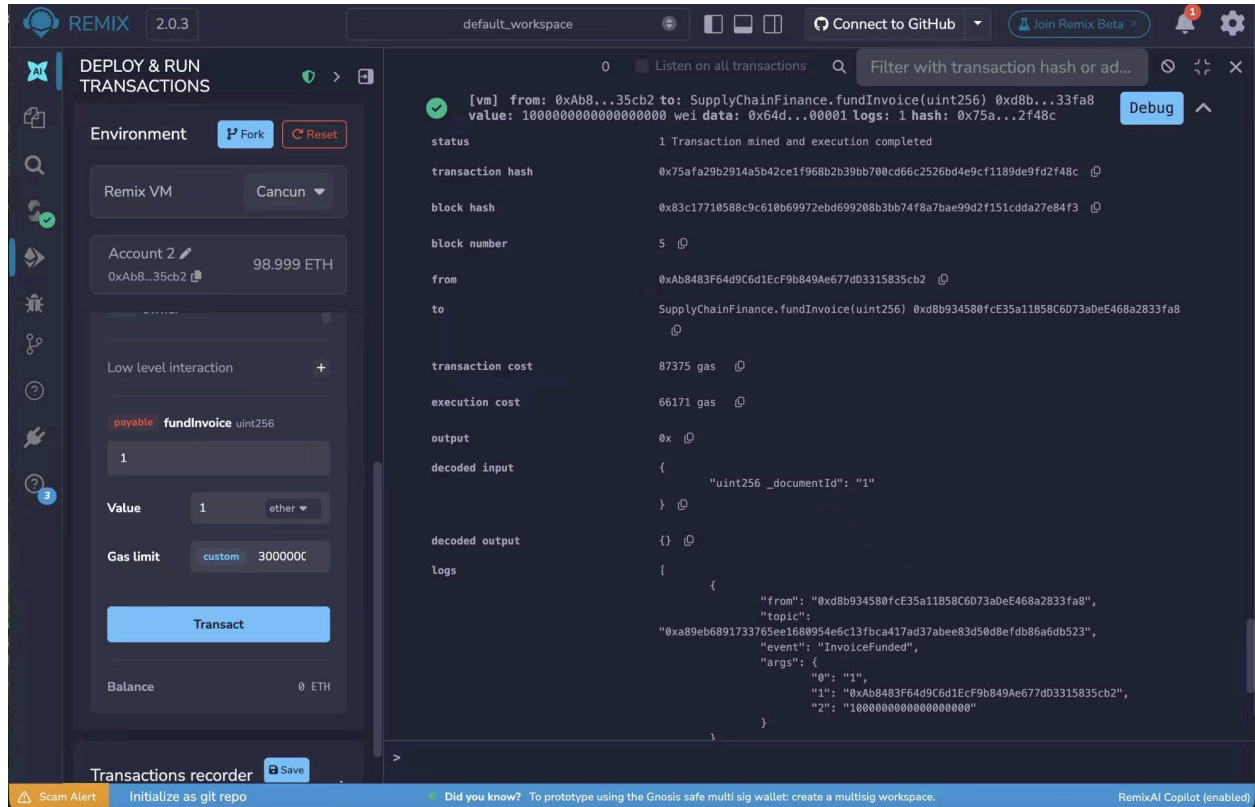


Figure 3: Fund Invoice

5. Test Case 4

Repay Invoice

- Action: Account 2 calls repayInvoice(1) with 1 ether
- Expected Result:
 - Document status changes to Repaid (3)
 - Lender receives repayment
- Evidence: Figure 4 (InvoiceRepaid event)

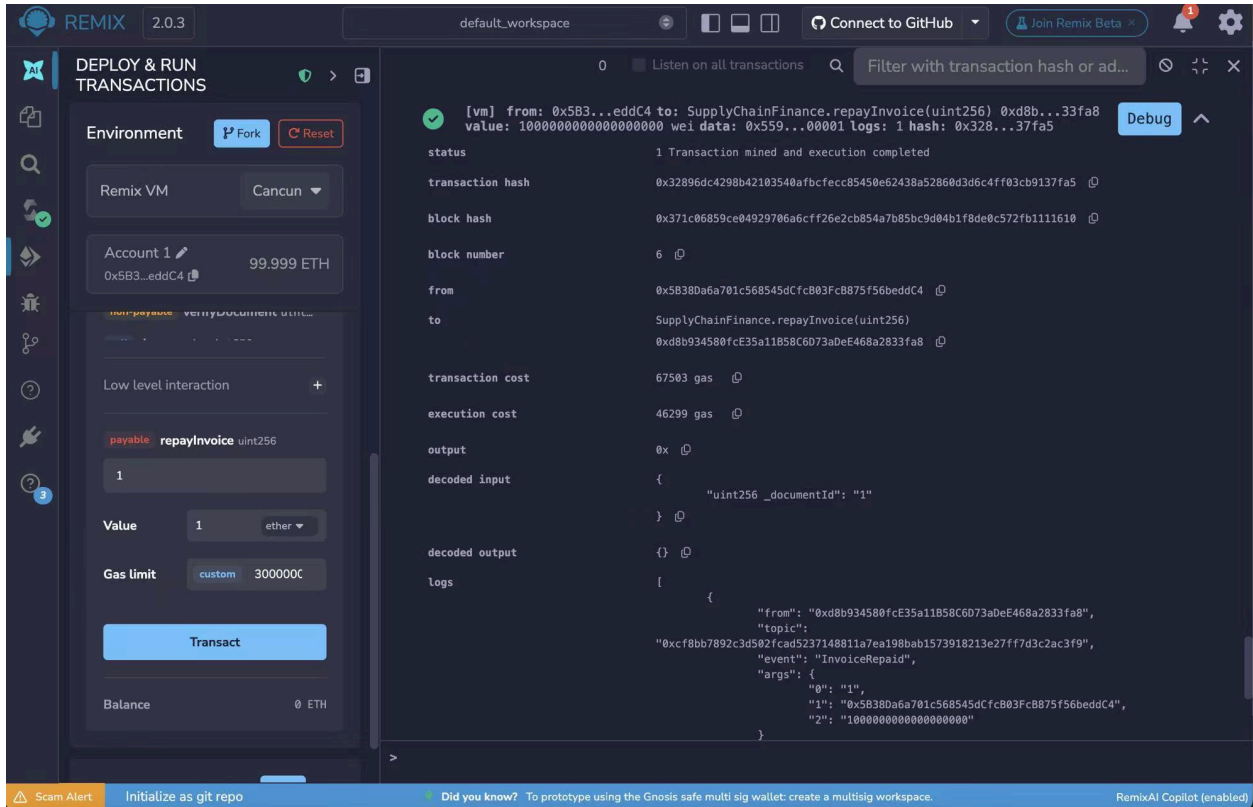


Figure 4: Repay Invoice

6. Test Case 5

Retrieve Document Data

- Action: Call `getDocument(1)`
- Expected Result:
 - Returns correct document details:
 - `documentId = 1`
 - `documentHash = "INV001"`
 - exporter, verifier, lender addresses
 - `invoiceAmount`, `fundedAmount`, `repaidAmount`
 - `status = Repaid`
- Evidence: Figure 5 (Decoded output)

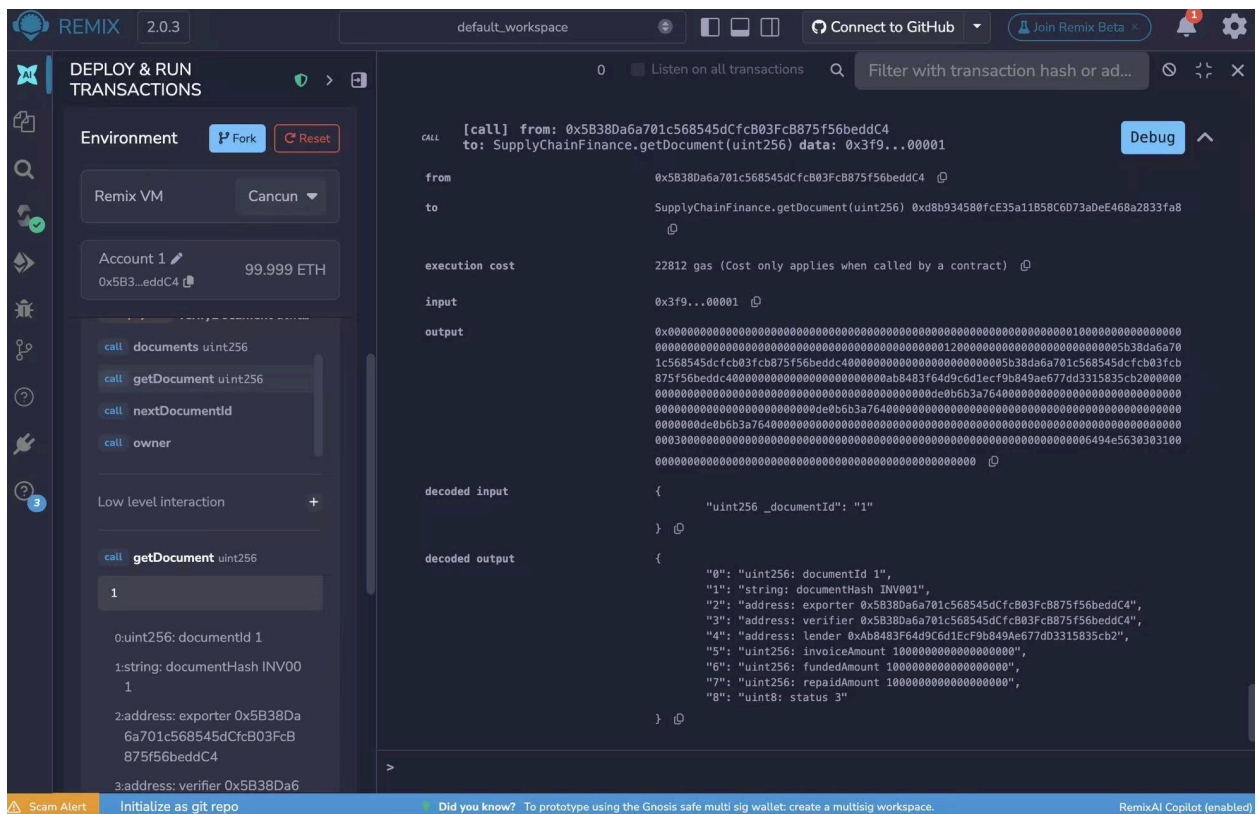


Figure 5: Retrieve Document Data

7. Test Case 6

Prevent Funding Before Verification

- Action: Try to call fundInvoice() before verifyDocument()
- Expected Result:
 - Transaction fails (invalid state)

8. Test Case 7

Prevent Duplicate Funding

- Action: Try to fund the same invoice again after it is already funded
- Expected Result:
 - Transaction fails